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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course (DSC B) - MG2DSCFIM100 - Introduction to Fitness Training and Management

(2024 ADMISSION ONWARDS)

Answer Key

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Part A

MCQ

Answer all questions

Each question carries 1 mark

1. Which component of fitness is MOST associated with heart and lung efficiency?

c). Cardiovascular endurance

2. What is the primary goal of strength and conditioning programs?

b). Enhance athletic performance and reduce injury risk

3. Which of the following is the most effective motivational strategy for a fitness class?

a). Creating a supportive, positive environment and recognizing individual achievements

4. What is the primary goal of client acquisition in fitness?

b). To build a loyal client base

5. What is a key factor to consider when analyzing financial performance for a personal training business?

b). Analyzing both income and expenses to ensure sustainable profitability

6. How does maintaining proper alignment during exercises help in injury prevention?

b). It evenly distributes forces across joints and muscles

7. A strong core helps improve:

a). Balance and coordination

8. How do fitness wearables help with health monitoring beyond tracking physical activity?

b). They provide insights into sleep patterns, heart rate variability, and stress levels

9. Which of the following is an important financial metric for personal trainers to monitor regularly?

b). Profit margin (Revenue - Costs)

10. When designing a fitness program for someone with a disability, which of the following is most important?

d). Consider the individual's specific abilities and limitations

[1x10=10]

Part B

Short Answer

Answer any 10 questions

Each question carries 2 marks

11. Name two well-known organizations that provide certification for personal trainers.

I. American Council on Exercise (ACE)

II. National Academy of Sports Medicine (NASM).

12. How does yoga help in improving flexibility?

`Yoga utilizes static and dynamic stretching techniques to increase joint range of motion and lengthen muscle fibers.

13. How does the principle of progression apply to sport-specific training?

It involves the gradual increase of training loads (intensity/volume) to ensure continuous adaptation and prevent plateaus.

14. What are the benefits of pursuing advanced certifications?

- Increased expertise
- higher professional credibility
- the ability to command higher rates or specialize in niche markets.

15. How do fitness wearables help track progress in physical activity?

They provide objective data (heart rate trends, step counts, sleep quality) that allow for long-term tracking of physiological improvements.

16. How does exercise science contribute to injury prevention?

By applying biomechanical principles to correct movement patterns and ensuring balanced muscle development.

17. Classify the basic principles of resistance training.

- Overload,
- Specificity
- Progression.

18. Why is it important to pay attention to participants' fatigue levels during a class?

Fatigue leads to compromised form, which significantly increases the risk of acute injury.

19. What role do online platforms play in networking and professional development?

They offer access to global research, networking with industry peers, and platforms for personal branding.

20. List down the best conditioning exercises for improving cardiovascular endurance.

- Running
- Swimming
- Cycling
- Rowing
- High-Intensity Interval Training (HIIT).

21. Why is it unethical for a fitness trainer to guarantee specific results to clients?

Because biological responses and client adherence are variable factors outside the trainer's absolute control.

22. What is the importance of flexibility in injury prevention?

- It allows joints to move through their full range of motion
- Preventing muscle strains during unexpected or intense movements.

[2x10=20]

Part C

Short Essay

Answer any 4 questions

Each question carries 5 marks

23. Explain how exercise can be beneficial for individuals with medical condition?

- Exercise helps improve heart health, blood circulation, and lung function in individuals with medical conditions.
- It helps control and manage diseases such as Type 2 Diabetes, high blood pressure, and obesity.
- Regular physical activity strengthens muscles and joints, reducing pain and improving mobility in some conditions.
- Exercise also improves mental health by reducing stress, anxiety, and depression.
- With proper guidance, exercise can improve overall quality of life, energy levels, and independence in daily activities.

24. Why is it important for fitness professionals to post consistent content on social media, and how can consistency lead to increased visibility and client acquisition?

- Posting consistent content helps fitness professionals build a strong online presence and personal brand.
- Regular posts keep followers engaged and increase trust and credibility among potential clients.
- Consistency improves visibility because social media platforms often promote active accounts to more users.
- Sharing useful fitness tips, workouts, and success stories attracts new audiences and potential clients.
- Consistent content helps maintain audience interest, which can lead to higher client inquiries and business growth.

25. Describe three key functional exercises and explain how they improve movement efficiency.

1. Squat: Improves daily sitting and standing mechanics.
2. Deadlift: Develops safe lifting habits for external loads.
3. Lunge: Enhances single-leg stability and gait efficiency.

26. Explain the importance of effective communication between a trainer and a client.

- Effective communication helps build trust and understanding between the trainer and the client.
- It allows trainers to clearly explain exercise techniques, goals, and safety instructions.
- Good communication helps trainers understand the client's needs, fitness level, and limitations.
- It increases the client's motivation, confidence, and commitment to training programs.
- Effective communication reduces misunderstandings and helps achieve better performance and fitness results.

27. Explain the difference between continuous training and interval training in endurance development.

- Continuous training is performed at a steady pace without rest, while interval training includes periods of exercise alternated with rest or recovery.
- Continuous training mainly develops aerobic endurance, whereas interval training improves both aerobic and anaerobic endurance.
- In continuous training, the intensity remains constant, but interval training uses varying intensities from high to low.
- Examples of continuous training are jogging and cycling, while interval training includes sprints and HIIT workouts.
- Continuous training builds stamina over longer durations, while interval training improves speed, power, and recovery ability more quickly.

28. Examine the relationship between flexibility and injury prevention in sports training

- Flexibility improves range of motion and helps athletes perform movements efficiently without putting excessive strain on muscles and joints.
- Good flexibility reduces muscle tightness and stiffness, lowering the chances of muscle strains and cramps during sports activities.
- Proper flexibility supports correct body mechanics and posture, which helps prevent overuse injuries and stress on joints.
- Dynamic stretching before exercise prepares muscles for activity and can reduce the risk of injuries during performance.
- Excessive flexibility without strength can be harmful, as it may cause joint instability and increase injury risk.
- Flexibility alone cannot prevent injuries; strength training, proper technique, rest, and conditioning are also important factors.

- Different sports require different levels of flexibility, so athletes should focus on sport-specific mobility and stretching exercises.

[5x4=20]